

NAVEEN MYSORE

nmysore.work@gmail.com

San Francisco Bay Area

Senior Research Engineer | Senior Software Engineer

H-1B visa holder with approved I-140

Current: Dyssonance AI (2026-present)

Previously: UCSB (2024-2026) | Salesforce/Tableau (2019-2024) | Dell EMC (2015-2019)

linkedin.com/in/mysore-naveen | github.com/NAVEENMN | naveenmn.github.io | Google Scholar

Professional Summary

Software Engineering: Seasoned engineer with 10+ years of combined software engineering and research experience across Salesforce/Tableau, Dell EMC, UCSB, and Dyssonance AI. Translate requirements into system designs and product capabilities, coordinate delivery across teams, and use systematic investigation to diagnose bottlenecks and improve performance. Experience spans distributed systems, enterprise data infrastructure, security features, and ML experimentation platforms.

AI Research and Scientific Computing: Research focused on interpretable and explainable AI, reinforcement learning, causal structure discovery, and language-model memory and reasoning. Use hypothesis-driven experiments to investigate Markov-property violations in reinforcement learning and physical parameter recovery from distributed observations. Build AI-assisted workflows for scientific discovery, automating hypothesis testing, experimentation, and result analysis. Creator of InKAN, an open-source PyTorch library for interpretable neural networks; author of an accepted RLC 2026 paper and scientific computing preprints; and named inventor on a U.S. provisional patent application.

Technical Skills

Area	Skills and tools
AI, ML & Research	Machine learning, deep learning, large language models (LLMs), reinforcement learning, fine-tuning (SFT, RLHF, PPO, GRPO), reasoning, reward modeling, graph neural networks, explainable AI, causal inference, retrieval-augmented generation (RAG)
Software Engineering	Distributed systems, system design, software architecture, requirements analysis, API development, performance optimization, testing, observability, root cause analysis
Scientific Computing	Numerical methods, statistical modeling, hypothesis testing, time series analysis, uncertainty quantification, physical parameter estimation, KANs, B-splines
Languages and Libraries	Python, PyTorch, Hugging Face Transformers, TRL, DeepSpeed, NumPy, SciPy, scikit-learn, Weights & Biases (W&B)
ML Systems	Distributed training, multi-GPU training, model optimization, model deployment, inference optimization, ML pipelines, experiment tracking, hyperparameter tuning, model evaluation
Cloud and Data	AWS, Google Cloud (GCP), Terraform, Apache Kafka, PostgreSQL, vector databases

Professional Experience

Dyssonance AI (stealth)

Feb 2026 - Present

Senior Research Engineer

Develop Llama-based memory and reasoning models through joint training with graph neural network outputs. Train reasoning models for long-horizon planning and convert research prototypes into product features.

University of California, Santa Barbara

Oct 2024 - Feb 8, 2026

Research Specialist | Prof. Yao Qin's team, REAL AI

Built AI-assisted experiment loops to explore hypotheses, run hyperparameter scans, and review results, reducing setup from hours to minutes and increasing experiment throughput **5x**. Automated Weights & Biases (W&B) setup and result alerts, stored results in PostgreSQL, and provided scripts to retrieve results and plot curves for weekly research reviews. Built SFT-to-PPO workflows, fine-tuned LLaMA-3.1 8B for nutrition reasoning with streaming human feedback for continual RLHF, and deployed inference on AWS. Supported ML engineering across research teams and developed Nutribench to collect health data for AI training.

Research code: github.com/ucsb/markovianess

Code: github.com/naveen-mysore/Reinft | Platform demo | Nutribench app

Professional Experience Continued

Salesforce | San Francisco, CA

Feb 2019 - Oct 2024

Senior Software Engineer | Nishi Bhonsle's Core Build and Tableau Extract team

Improved Tableau's data extraction pipeline performance by **5%** through bottleneck analysis and optimization under heavy loads. Led cross-team delivery of Customer Managed Encryption Keys (CMEK), enabling secure onboarding of McKinsey. Built Kafka-based adoption tracking, A/B testing dashboards, a statistical hypothesis testing platform, and AI-assisted intelligent metrics used by cross-functional teams.

Dell EMC | Santa Clara, CA

Jun 2015 - Feb 2019

Software Engineer

Designed a distributed data-server synchronization algorithm using balanced N-ary tree updates to reduce inconsistencies. Benchmarked server workloads on AWS and used regression modeling to match workloads to instance types for cost-performance efficiency.

Open Source Projects

InKAN | Author and Maintainer

Developed and published InKAN, a PyTorch library for B-spline Kolmogorov-Arnold Networks supporting CPU, CUDA, and Apple Silicon, with reusable layers and visualization tools. Optimized B-spline evaluation with torch.compile kernel fusion, achieving 2.8-3.5x faster forward passes than Cox-de Boor recursion in tested NVIDIA H100 benchmarks.

pypi.org/project/inkan | github.com/NAVEENMN/inkan

Selected Publications and Preprints

Prediction-Based Markov Violation Scores for Detecting Non-Markovian Observations in Reinforcement Learning

Naveen Mysore | Reinforcement Learning Journal, vol. 7, 2026; accepted at RLC 2026.

rlj.cs.umass.edu/2026/papers/Paper70.html

Recovering Physical Parameters from Fragmented Observations via Exact Distributed Spline Merging

Naveen Mysore | arXiv preprint, 2026.

arxiv.org/abs/2609.16579

* Recovered diffusion coefficients and wave speed with 0.11% and 0.12% error in synthetic experiments; validated distributed reconstruction on 41 years of NOAA data.

InKAN: B-Spline KANs via Truncated Power Form

Naveen Mysore | arXiv preprint, 2026.

arxiv.org/abs/2609.01956

Patent Application

System and Method for Distributed Spatiotemporal Field Reconstruction and Physical Parameter Recovery Using Composable Spline Sufficient Statistics

Named inventor | U.S. provisional application 64/157,134

Education

Master's in Computer Science (Machine Learning)

2013 - 2015

University of North Carolina at Charlotte

Thesis: Elastic group recommendation systems for multivariate dynamic attributes. IEEE ICIS 2016.

Bachelor's in Electrical and Electronics Engineering

2009 - 2013

PES University (PESIT)